

Minimal audio kit — 7 parts

Your carrier PCB + seven purchased parts per kit. For ten kits: buy ten of each, preferably one 10-piece lot. Scope confirmed 9 September 2026: **joystick on a board with pins; encoder bare.**

#	Part	Required version	AliExpress candidate
1	ESP32-S3 SuperMini	ESP32-S3FH4R2, 4 MB flash / 2 MB PSRAM, USB-C	1–10 pcs
2	PCM5102A I2S DAC	Purple board , with audio jack	1–10 pcs
3	Analog joystick	KY-023-style board with fitted pins and cap ; X, Y and push button	1/5/10 pcs
4	Rotary encoder	Bare EC11 with push button , vertical shaft	10 pcs
5	I2S microphone	Small round six-pin board ; INMP441 assumed from this project	1/5/10 pcs
6	0.96-inch OLED	SSD1306, 128×64, I2C SDA/SCL , four pins	10 pcs
7	Gyroscope / accelerometer	MPU6050 GY-521 board , I2C	10 pcs

Ordering status: these are listing leads. Select the 10-piece variant where available. Free shipping/Choice, fitted headers, the purple DAC version, round microphone version and exact board drawings remain **unverified**. Country/postcode and readable seller details are still needed. The ten-kit batch is **70 purchased parts + 10 of your carrier PCBs**.

Documentation and CAD: [PCM5102A datasheet](#) · [INMP441 datasheet](#) · [MPU6050 datasheet](#) · [KY-023 module manual/dimensions reference](#) · [SSD1306 OLED module dimensions reference](#). Reference-board dimensions must be matched to the actual supplier's boards. Exact DipTrace libraries remain unverified; existing drawings and CAD leads are in the [detailed research](#).

Carrier design: use the actual module header layouts and bare encoder footprint. Keep joystick outputs and I2C pull-ups at 3.3 V. The purple DAC's input power and configuration straps depend on its revision. Its output is line level; if headphone drive is wanted, account for that in the carrier design. [TI DAC specification](#).

Listing-title evidence: [ESP32](#) · [DAC](#) · [joystick](#) · [encoder](#) · [mic](#) · [OLED](#) · [IMU](#). These indexed records do not verify shipping or the selected variant.

[Editable seven-part CSV](#) · [Earlier expanded research](#)